

Homework 2

● Graded

Student

Shani De Silva

Total Points

100 / 100 pts

Question 1

1.1 Question 1a

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Mistake in the answer to 1 out of the 5 cases.

- 2 pts Mistake in the answer to 2 out of the 5 cases.

- 3 pts Mistake in the answer to 3 out of the 5 cases.

- 4 pts Mistake in the answer to 4 out of the 5 cases.

- 5 pts Mistake in the answer to 5 out of the 5 cases.

- 5 pts No answer is written.

1.2 Question 1b

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Mistake in the answer to 1 out of the 5 cases.

- 2 pts Mistake in the answer to 2 out of the 5 cases.

- 3 pts Mistake in the answer to 3 out of the 5 cases.

- 4 pts Mistake in the answer to 4 out of the 5 cases.

- 5 pts Mistake in the answer to 5 out of the 5 cases.

- 5 pts No answer is written.

1.3 Question 1c

4 / 4 pts

✓ - 0 pts Correct

- 4 pts Answer is wrong or no answer is written.

- 1 pt Incorrect carryover answer from 1b but good explanation

1.4 Question 1d

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Answer is partially correct.

- 3 pts Answer is completely wrong or no answer is written.

1.5 Question 1e

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Answer is partially correct.

- 3 pts Answer is completely wrong or no answer is written.

- 1 pt No example given

Question 2

2.1 Question 2a

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Answer is correct, but the methodology description is wrong or imprecise.

- 2 pts Methodology description is correct, but the answer is wrong.

- 4 pts Both the methodology description and the answer are wrong.

- 4 pts No answer is written.

2.2 Question 2b

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation of K is wrong.

- 1 pt Calculation of I_D is wrong.

- 2 pts Comparison against the measured value is wrong.

- 4 pts No answer is written.

- 1 pt Carry over estimation error but correct methodology

2.3 Question 2c

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation of K_v is wrong.

- 1 pt Calculation of I_D is wrong.

- 2 pts Comparison against the measured value is wrong.

- 4 pts No answer is written.

- 1 pt Carry over error but methodology is correct

2.4 Question 2d

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Answer is partially correct.

- 4 pts Answer is completely wrong or no answer is written.

2.5 Question 2e

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Answer is partially correct.

- 4 pts Answer is completely wrong or no answer is written.

Question 3

3.1 Question 3a

5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts Answer is partially correct.

- 5 pts Answer is completely wrong or no answer is written.

3.2 Question 3b

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Method for calculating r is correct, but the answer is wrong.

- 2 pts The calculated value of r is correct, but the method is wrong.

- 4 pts Both the method and the calculated value of r are wrong.

- 4 pts No answer is written.

3.3 Question 3c

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation for $r = 1$ is wrong.

- 1 pt Calculation for $r = 2$ is wrong.

- 1 pt Calculation for $r = 3$ is wrong.

- 1 pt Explanation of what happens to VTC with an increase in W_p/W_n is wrong.

- 4 pts No answer is written.

3.4 Question 3d

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Answer is partially correct.

- 3 pts Answer is completely wrong or no answer is written.

3.5 Question 3e

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Prediction is correct, but the explanation is wrong.

- 2 pts Explanation is correct, but the prediction is wrong.

- 4 pts Both the explanation and the prediction are wrong.

- 4 pts No answer is written.

Question 4

4.1 Question 4a

5 / 5 pts

✓ - 0 pts Correct

- 1.5 pts Calculation for design A is wrong.

- 1.5 pts Calculation for design B is wrong.

- 1.5 pts Calculation for design C is wrong.

- 5 pts No answer is written.

4.2 Question 4b

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation for design A is wrong.

- 1 pt Calculation for design B is wrong.

- 1 pt Calculation for design C is wrong.

- 4 pts No answer is written.

4.3 Question 4c

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation for design A is wrong.

- 1 pt Calculation for design B is wrong.

- 1 pt Calculation for design C is wrong.

- 4 pts No answer is written.

4.4 Question 4d

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation for design A is wrong.

- 1 pt Calculation for design B is wrong.

- 1 pt Calculation for design C is wrong.

- 1 pt Discussion about the smallest V_M spread and/or the switching threshold is wrong.

- 4 pts No answer is written.

- 0.5 pts Missing smallest V_M spread answer.

- 1 pt Discussion about the smallest V_M spread and/or the switching threshold is missing.

- 0.5 pts Smallest V_M spread answer is wrong.

✓ - 0 pts Correct

- 1.5 pts Answer is partially correct.

- 3 pts Answer is completely wrong or no answer is written.

Question 5

5.1 Question 5a

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Calculation in the cutoff region is wrong.

- 1 pt Calculation in the linear region is wrong.

- 1 pt Calculation in the saturation region is wrong.

- 0.5 pts Capacitance split between source and drain for linear region is wrong.

- 0.5 pts Capacitance split between source and drain for saturation region is wrong.

- 4 pts No answer is written.

5.2 Question 5b

4 / 4 pts

✓ - 0 pts Correct

- 1 pt C_{GSO} is calculated wrong.

- 1 pt C_{GDO} is calculated wrong.

- 1 pt Fringe capacitance is wrong.

- 4 pts No answer is written.

5.3 Question 5c

4 / 4 pts

✓ - 0 pts Correct

- 2 pts C_{diff} is wrong.

- 2 pts Source-plus-drain diffusion capacitance is wrong.

- 4 pts No answer is written.

5.4 Question 5d

5 / 5 pts

✓ - 0 pts Correct

- 2.5 pts Calculated C_{GDO} is wrong.

- 2.5 pts Comparison with physical C_{GDO} is wrong.

- 5 pts No answer is written.

5.5 Question 5e

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Answer is partially correct.

- 3 pts Answer is completely wrong or no answer is written.

Q1 Please see Printed Pg 1 for Working & calculations

Case	V_{gs}	V_{ds}	V_{gt}
1	0.25	0.8	-0.1
2	0.55	0.1	0.2
3	0.7	0.4	0.35
4	0.9	0.4	0.55
5	1	0.8	0.65

$$V_t = 0.35$$

$$V_{dd} = 1$$

$$V_{gt} = V_{gs} - V_{ds}$$

Q1a Long-channel model $\rightarrow V_{dsat} = V_{gt}$.

Case	V_{gt}	V_{ds}	V_{dsat}	Region of operation.
1	-0.1	0.8	-0.1	OFF. $V_{gt} < 0$.
2	0.2	0.1	0.2	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
3	0.35	0.4	0.35	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.
4	0.55	0.4	0.55	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
5	0.65	0.8	0.65	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.

Q1b Short-channel model $\rightarrow V_{dsat} = \min(V_{gt}, 0.45)$

case	V_{gt}	V_{ds}	V_{dsat}	Region of operation.
1	-0.1	0.8	-0.1	Off. $V_{gt} < 0$.
2	0.2	0.1	0.2	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
3	0.35	0.4	0.35	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.
4	0.55	0.4	0.45	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
5	0.65	0.8	0.45	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.

Q1c for the information specified, there are no cases where the long-channel & short channel models give different operating-region classifications. There are no disagreements to explain.

Q1d Early velocity saturation changes the intuition that drain current increases quadratically with V_{gs} because while in long-channel transistors where saturation happens through pinch-off short-channel transistors are saturated far earlier (normally due to pinch off but now due to induced electric fields causing saturation velocity v_{to}).

Case	Vgs (V)	Vds (V)	Vgt (V)
1	0.25	0.8	-0.1
2	0.55	0.1	0.2
3	0.7	0.4	0.35
4	0.9	0.4	0.55
5	1	0.8	0.65

Vt (V)	0.35
Vdd (V)	1

Long Channel Model		Vdsat = Vgt		
Case	Vgt	Vds (V)	Vdsat	Reg. of. Op.
1	-0.1	0.8	-0.1	Vgt<0, OFF
2	0.2	0.1	0.2	Vgt>0, Vds<Vdsat. LINEAR
3	0.35	0.4	0.35	Vgt>0, Vds>Vdsat. SATURATED
4	0.55	0.4	0.55	Vgt>0, Vds<Vdsat. LINEAR
5	0.65	0.8	0.65	Vgt>0, Vds>Vdsat. SATURATED

Short-Channel Model		Vdsat = MIN(Vgt, 0.45V)		
Case	Vgt	Vds (V)	Vdsat	Reg. of. Op.
1	-0.1	0.8	-0.1	Vgt<0, OFF
2	0.2	0.1	0.2	Vgt>0, Vds<Vdsat. LINEAR
3	0.35	0.4	0.35	Vgt>0, Vds>Vdsat. SATURATED
4	0.55	0.4	0.45	Vgt>0, Vds<Vdsat. LINEAR
5	0.65	0.8	0.45	Vgt>0, Vds>Vdsat. SATURATED

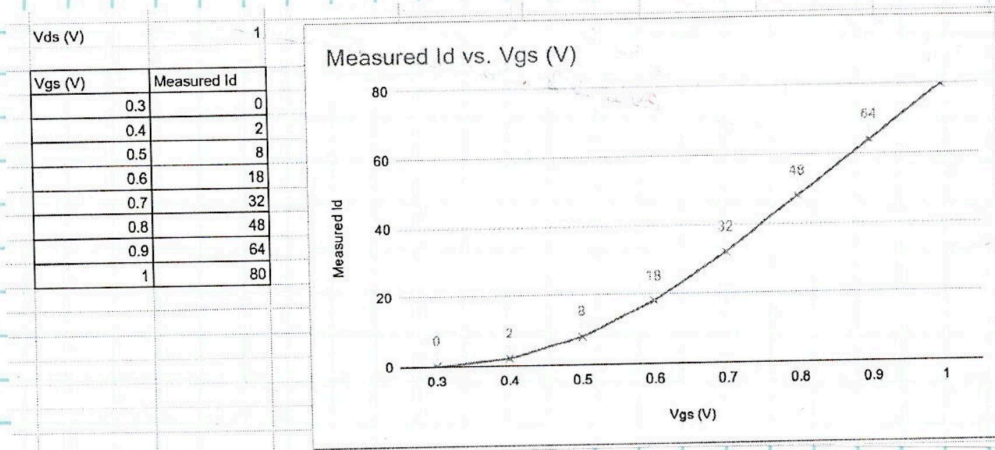
Page 1 Working
for Q1.

Q1D, cont) ~~at~~ for the carrier electrons far before any pinch-off can start to affect the drain current.

1e) This is not true: If V_{gs} is less than $V_{thres, on}$, despite any large V_{ds} value, the gate will remain OFF and therefore not saturated.

Example: Case 1. V_{ds} is 0.8 and yet the gate remains off and not saturated.

2a) After graphing V_{gs} and I_D , the V_T threshold is where I_D first equals 0. This is ~~at~~ 0.3 V.



(doesn't require a graph but I liked the visual reference for future Question 2 parts.

2b

Computed K at $V_{gs}=0.6$

↳ 200.

Predicted I_D at $V_{gs}=1$

↳ 98.

Actual I_D at $V_{gs}=1$

↳ 80.

Predicted > Actual.

Estimated V_t (V):	0.3	$I_D = K(V_{gs}-V_t)^2$ $K = I_D/(V_{gs}-V_t)^2$	
V_{gs}	Estimated V_t	I_D	K
0.6	0.3	18	200
1	0.3	98	200
1	0.3	80	163.2653061

Bold=Calculated

< using calculated K

< Actual values

Computed K :	200	QUADRATIC MODEL
Predicted I_D at $V_{gs}=1$:	98	
Actual I_D at $V_{gs}=1$:	80	

Prediction I_D is higher than measured value, and Prediction K is also higher than measured value.

2c

Computed K_v at $V_{gs}=0.8$

↳ 96.

Predicted I_D at $V_{gs}=1.0$

↳ 67.2

Actual I_D at $V_{gs}=1$

↳ 80.

Predicted < actual.

Estimated V_t (V):	0.3	$I_D = K_v(V_{gs}-V_t)$ $K_v = I_D/(V_{gs}-V_t)$	
V_{gs}	Estimated V_t	I_D	K
0.8	0.3	48	96
1	0.3	67.2	96
1	0.3	80	114.2857143

Bold=Calculated

< using calculated K

< Actual values

Computed K_v :	96	LINEAR MODEL
Predicted I_D at $V_{gs}=1$:	67.2	
Actual I_D at $V_{gs}=1$:	80	

Prediction I_D is lower than measured value, and Prediction K is also lower than measured value.

2d

Comparing the two models from 2B & 2C.

Quadratic model:
Prediction I_D is higher than measured value, and Prediction K is also higher than measured value.
Linear model:
Prediction I_D is lower than measured value, and Prediction K is also lower than measured value.

A linear model fits better as saturation is reached.

V_{gs}	I_D	
0.3	0	
0.4	2	+2
0.5	8	+6
0.6	18	+10
0.7	32	+14
0.8	48	+16
0.9	64	+16
1.0	80	+16

quadratic @ lower V_{gs}

linear @ higher V_{gs} .

(2e) estimated quadratic relationship:

$$I_D = K(V_{GT} - V_T)^2 \rightarrow I_D \propto V^2.$$

↑
Proportional
to

Propagation Delay (Time units) = τ tau?

~~$$\tau = \frac{C \times \Delta V}{I}$$~~

$\tau = \frac{C \times \Delta V}{I}$. If $I \propto V^2$, as V improves, I improves
By far more, so τ is not as big as
the developer expects.

~~Bad~~

$$V_m = \frac{V_{DD}}{2 + r}$$

$$(3a) \quad I_{DN} = I_{DP} \quad V_{TN} = |V_{TP}| = 0.3V. \quad V_{DD} = 1V$$

$$K_n W_n (V_M - V_{TN}) = K_p W_p (V_{DD} - V_M - |V_{TP}|)$$

$$\frac{K_n}{K_p} (V_M - V_{TN}) = \frac{W_p}{W_n} (V_{DD} - V_M - |V_{TP}|)$$

$$2(V_M - V_{TN}) = r(V_{DD} - V_M - |V_{TP}|)$$

$$2V_M - 2(0.3) = rV_{DD} - rV_M - 0.3r$$

$$2V_M + rV_M = r(1) - 0.3r + 0.6$$

$$(2+r)V_M = 0.7r + 0.6$$

$$V_M = \frac{0.7r + 0.6}{2+r}$$

$$(3b) \quad V_M = V_{DD}/2 = 1/2$$

$$0.5 = \frac{0.7r + 0.6}{2+r}$$

$$(2+r)(0.5) = 0.7r + 0.6$$

$$1 + \frac{r}{2} = 0.7r + 0.6$$

$$0.4 = 0.2r$$

$$4 = 2r$$

$$2 = r$$

$$(3c) \quad V_M = \frac{0.7r + 0.6}{2+r}$$

$$\frac{W_P}{W_N} = R$$

$$r=1, \quad V_M = \frac{0.7+0.6}{3} = \frac{1.3}{3} = \frac{13}{30} V.$$

$$r=2, \quad V_M = \frac{1.4+0.6}{4} = \frac{2}{4} = \frac{1}{2} V.$$

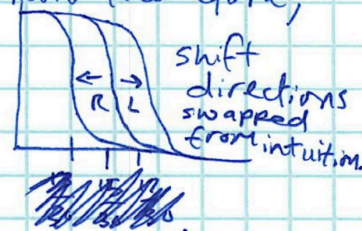
← 15/30 for ease of comparison.

$$r=4, \quad V_M = \frac{2.8+0.6}{2+4} = \frac{3.4}{6} = \frac{34}{60} = \frac{17}{30} V.$$

Increasing R , $\frac{W_P}{W_N}$, makes the PMOS side of the

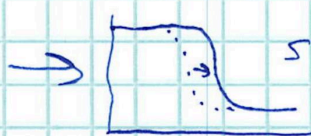
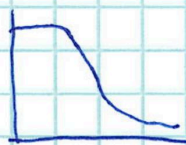
inverter stronger. This means that the transfer curve towards the PMOS side, where the output approaches zero, which is off to the ~~right~~ left now. Therefore, shift right.

electronics.stackexchange.com/questions/593213/vtc-curve-and-the-delay-time-of-cmos-inverter



~~3d~~ If NMOS is stronger than PMOS*, it stays dominant for longer, so is shifted left.

(3e)



shift left.

*than intended,

(3d)

gain ~~over~~ magnitude = amount the ~~in~~ inverter amplifies the signal away from V_M . $N_{ML} = V_{IL} - V_{OL}$, $N_{MH} = V_{IH} - V_{OH}$. To improve N_{ML} , where the input-low noise affects, choose a higher V_M to increase V_{IL} .

4a

Vdd (V)

1

 $I_{dn} = I_{dp}$

Vtn (V)

0.3

 $K_n/K_p = 2$ $|V_{tp}|$ (V)

0.3

 $W_p/W_n = r$

$$I_{dn} = k_n W_n (V_m - V_{tn})$$

$$I_{dp} = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$2 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$2 V_m - 0.6 = r (0.7 - V_m)$$

$$2 V_m - 0.6 = 0.7r - r V_m$$

$$2 V_m + r V_m = 0.7r + 0.6$$

$$(2+r) V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (2+r)$$

Design	Wn	Wp	r	Vm
A	1	1	1	0.4333333333
B	1	2	2	0.5
C	1	3	3	0.54

$$\frac{K_n}{K_p} = 2$$

4b

Vdd (V)

1

 $I_{dn} = I_{dp}$

Vtn (V)

0.3

 $K_n/K_p = 1.6$ $|V_{tp}|$ (V)

0.3

 $W_p/W_n = r$

$$I_{dn} = k_n W_n (V_m - V_{tn})$$

$$I_{dp} = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$1.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$1.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$1.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$1.6 V_m - 0.6 = r (0.7 - V_m)$$

$$1.6 V_m - 0.6 = 0.7r - r V_m$$

$$1.6 V_m + r V_m = 0.7r + 0.6$$

$$(1.6+r) V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (1.6+r)$$

Design	Wn	Wp	r	Vm
A	1	1	1	0.5
B	1	2	2	0.5555555556
C	1	3	3	0.5869565217

$$\frac{K_n}{K_p} = 1.6$$

changed derivation
to show working

$$\frac{K_n}{K_p} = 1.6$$

derivation
typed on
google
sheets
for legibility

4c

Vdd (V)	1	I _{dn} = I _{dp}
V _{tn} (V)	0.3	K _n /K _p = 2.6
V _{tp} (V)	0.3	W _p /W _n = r
I _{dn} = k _n W _n (V _m - V _{tn})		I _{dp} = K _p W _p (V _{dd} - V _m - V _{tp})

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$2.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$2.6 V_m - 0.6 = r (0.7 - V_m)$$

$$2.6 V_m - 0.6 = 0.7r - rV_m$$

$$2.6 V_m + rV_m = 0.7r + 0.6$$

$$(2.6 + r)V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (2.6 + r)$$

$$\frac{K_n}{K_p} = 2.6$$

for all taped-in grids,
used formula to
calculate on google sheets
to avoid copying errors

Design	W _n	W _p	r	V _m
A	1	1	1	0.3611111111
B	1	2	2	0.4347826087
C	1	3	3	0.4821428571

4d

Robustness. $\Delta V_m = V_{m \text{ MAX}} - V_{m \text{ MIN}}$

Design	Nominal	Slow Nmos Fast Pmos	Fast Nmos Slow Pmos	V _m Spread
A	0.4333333333	0.5	0.3611111111	0.1388888889
B	0.5	0.5555555556	0.4347826087	0.1207729469
C	0.54	0.5869565217	0.4821428571	0.1048136646

Design C has the smallest V_m spread

V_{dd}/2: 0.5

Design B is closest to V_{dd}/2 under nominal conditions.

4e

For a noisy environment, need to maximise noise margins to ensure a larger range of values that aren't too close to point where $V_{in} = V_{out}$.

low V_m spread for higher reliability when noisy. despite how heavily Design C is weighted area-wise, due to the size of the PMOS, the size was not mentioned as a limiting factor and the increased size makes it more resistant to manufacturing variability ("sensitivity to process variation")
Would choose C!

5a C_{gc} in cutoff, linear, saturation.

"gate channel capacitance is approximately zero in cutoff" - homework bullet point.

$$C_{gc}(\text{cutoff}) \approx 0.$$

"In linear operation, total gate channel capacitance is $C_{ox}WL$ & is split evenly between source & drain" - HW.

$$C_{ox} = 20 \text{ fF}/\mu\text{m}^2. \quad W = 1 \mu\text{m}. \quad L = 0.1 \mu\text{m}.$$

$$C_{gc}(\text{Linear}) = C_{ox}WL = 20 \times 1 \times 0.1 = 2 \text{ fF}.$$

← split between source & drain equally.

"In saturation, total $[C_{gc}]$ is $\frac{2}{3} \times C_{ox}WL$ and is assigned to the source side." - HW.

$$C_{ox}WL = 2 \text{ fF}. \quad C_{gc} = \frac{2}{3} C_{ox}WL = \frac{4}{3} \text{ fF} \leftarrow \text{assigned to source side.}$$

5b $C_{gs0} = C_{gdo} = C_0 W = 0.25 \times 1 = 0.25.$

$C_0 = 0.25$ $W = 0.25$ $C_0 \uparrow$ $W \uparrow$

total fringe capacitance $\Rightarrow C_{\text{fringe, total}} = C_{\text{fringe}} W.$

$$C_{\text{fringe}} = 0.2 \text{ fF}. \quad W = 1 \mu\text{m}.$$

$$C_{\text{fringe total}} = 0.4$$

$C_{\text{fringe total}} = C_{gs0} + C_{gdo}$? unsure why else it would be calculated.

5c have to add diffusion capacitance.

$$C_{\text{diff}} = C_j L_s W + C_{jsw}(2L_s + W) + C_{jgate} W.$$

$$= 1 \times 0.15 \times 1 + 0.2(2 \times 0.15 + 1) + 0.1 \times 1$$

$$= 0.15 + 0.2(1.3) + 0.1$$

$$= 0.15 + 0.26 + 0.10 = 0.51 \leftarrow \text{diffusion capacitance per gate terminal.}$$

$$0.51 \times 2 = 1.02 \text{ fF: total source \& drain diffusion capacitance.}$$

ASK
YANG
WU!

5d)

Miller effect:

$$A_v = -5$$

$$C_{GDO \text{ effective}} \approx C_{GDO} (1 - A_v)$$

$$1 - (-5) = 6$$

$$C_{GDO \text{ effective}} \approx 6 C_{GDO}$$

$$6 \times C_{GDO} = 6 \times 0.25 = 1.5 \text{ pF}$$

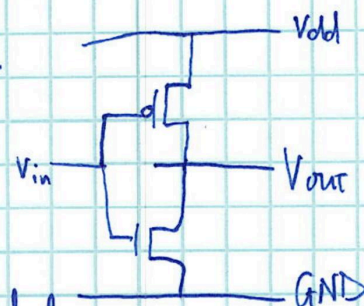
~~Physical C_{GDO}~~

ASK
YANG
WU!

Is it just that it contributes 1.5 pF to the input capacitance?? And that the effective contribution is 6x as big as the actual C_{GDO} ?

5e)

When the input changes from 0 to 1 or vice versa, the Miller effect is fully ignored by the student. The Miller effect is noticed during the changing of inputs and so by forgetting it the student underestimates in their predicted delay.



It is most noticeable when in VTC transition because that is where A_v is measured and that's where the biggest voltage change is.

AI STATEMENT: I didn't use AI, but I heavily used Yang Wu's teachings in his recitation.

